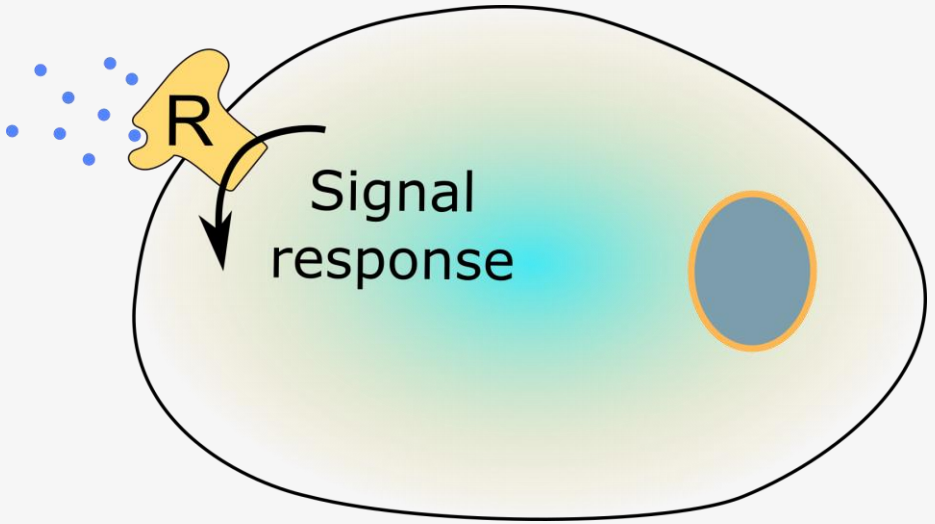
PHAR0002:
Drug Antagonism.

Dr. Guy Moss



Questions…

What is the highest efficacy an agonist can have, and would such a molecule be useful?

Can an artificial agonist ever have a higher efficacy than a natural agonist that produces a full tissue response?

Reminder – our updated model

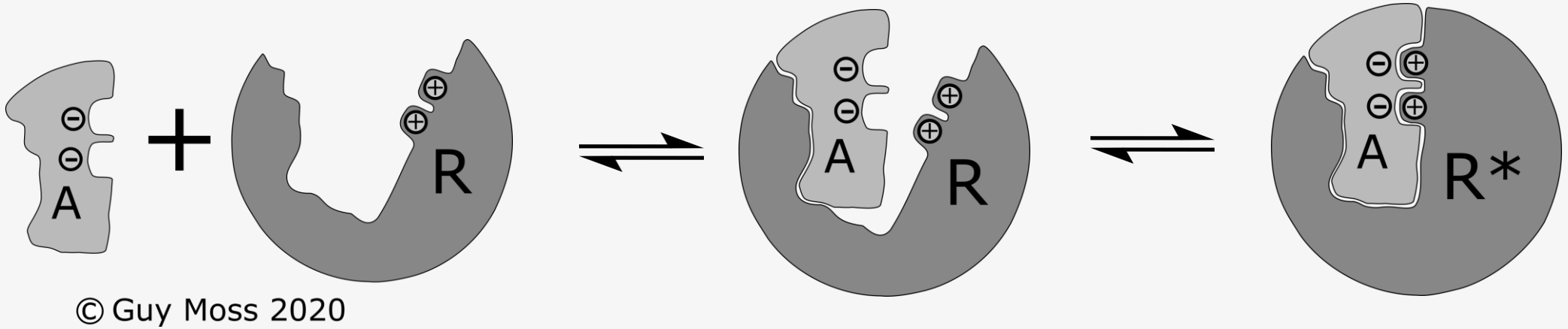
$$A + R \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} AR$$

$$A + R \overset{K_1}{\rightleftharpoons} AR \overset{E}{\rightleftharpoons} AR^*$$



The del Castillo-Katz Mechanism…

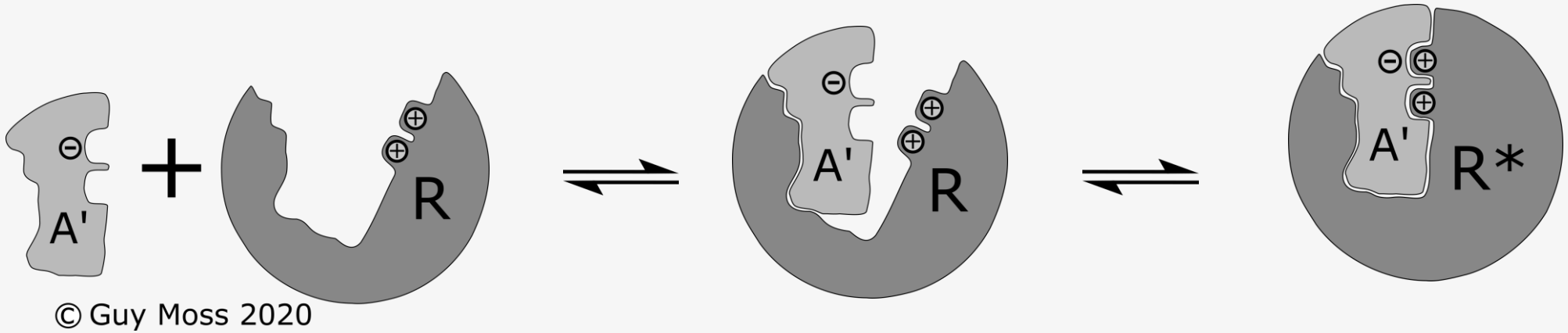
Bonds are the answer…



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Warning: **In what follows we are using charges as an aid to understanding the concepts of receptor-agonist interactions.** **Do not take this literally** **– receptor activation can occur via any bond type and charge interactions such as this are not meant to be a realistic representation of all (or indeed any) specific receptor activation process.**

What happens if we change the agonist?

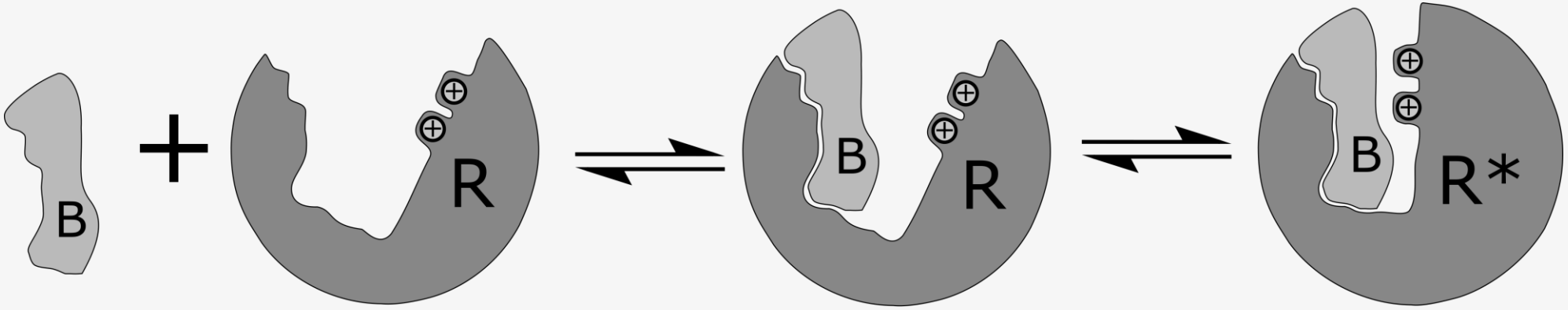


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Partial agonist

What if we change the agonist?



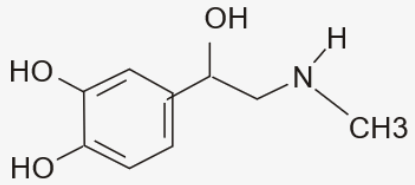


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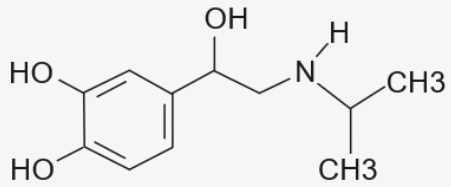
Reversible competitive antagonist

Adrenoceptor Ligands…

Adrenaline
(Epinephrine)

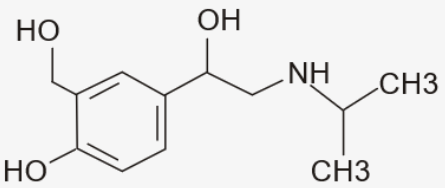


Isoprenaline



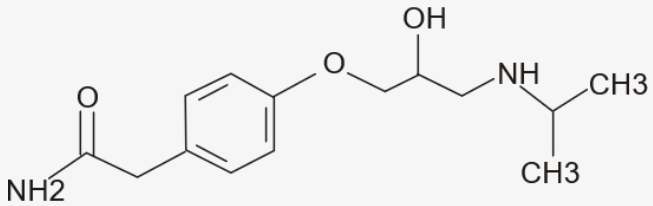
Non-selective
β-adrenoceptor agonist

Salbutamol



Selective β2-adrenoceptor
partial agonist

Atenolol



β2-adrenoceptor
antagonist

Drug antagonism

Antagonists are drugs that work to oppose the action of other chemical substances (e.g. agonists).

Most frequently antagonists work by binding to the same receptor macromolecule as the agonist.

Competitive antagonism

When both agonist and antagonist bind at the same receptor, one of the most common modes of antagonist action is competitive antagonism. This occurs when *the binding of agonist and antagonist is mutually exclusive.*

This may be because

1. The agonist and antagonist compete for the same binding site,

or

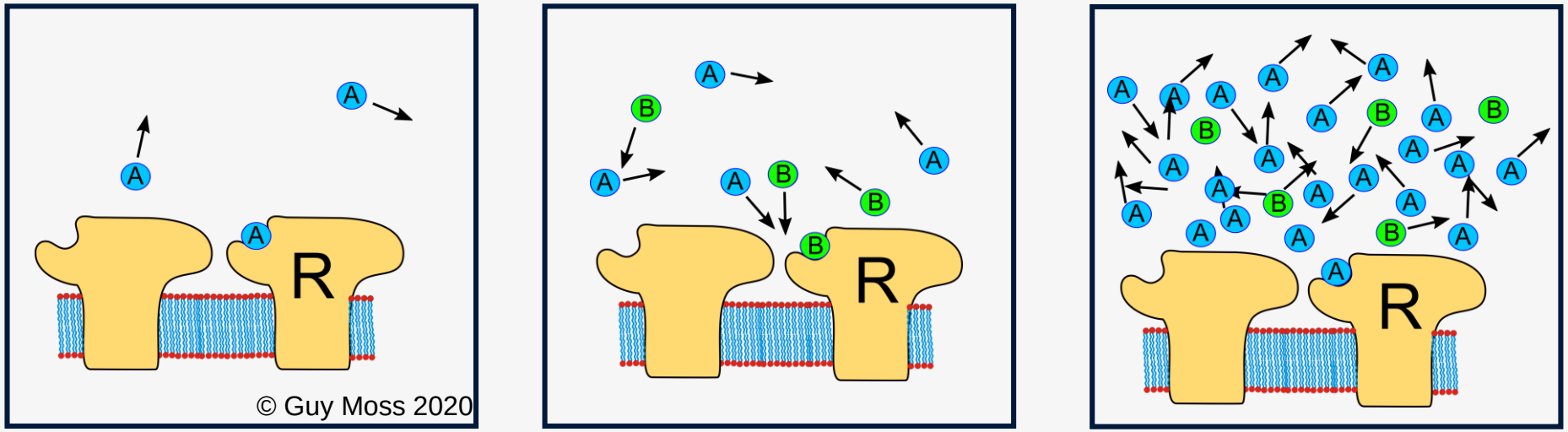
2. The agonist and antagonist combine with adjacent sites, which overlap,

or

3. Different sites are involved but that they influence the receptor macromolecule in such a way that agonist and antagonist molecules cannot be bound at the same time.

Competitive antagonists can be reversible or irreversible.

The nature of competition…



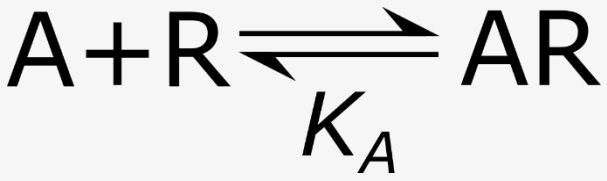
Our goal…

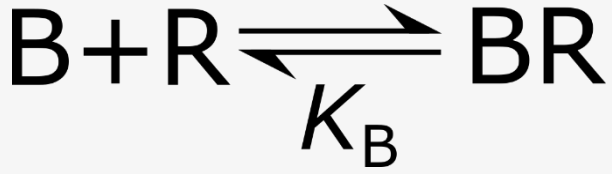
We want to examine dose-response curves and determine whether or not an antagonist works by reversible competitive inhibition.

Our first step…

Create a simple mathematical model to predict how the proportion of activated receptors will vary as we vary the concentrations of agonist ‘A’ and antagonist ‘B’.

A simplification…





Where K_A and K_B are the equilibrium dissociation constants associated with these reactions.

We now consider the population of all receptors…

$$A + R \underset{K_A}{\rightleftharpoons} AR$$

$$B + R \underset{K_B}{\rightleftharpoons} BR$$

$$[R]_T = [R] + [BR] + [AR] \quad \text{Eq. 1}$$

$$\frac{[R]_T}{[R]_T} = \frac{[R]}{[R]_T} + \frac{[BR]}{[R]_T} + \frac{[AR]}{[R]_T}$$

$$1 \quad p_R = \frac{[R]}{[R]_T} \quad p_{BR} = \frac{[BR]}{[R]_T} \quad p_{AR} = \frac{[AR]}{[R]_T}$$

$$1 = p_R + p_{BR} + p_{AR} \quad \text{Eq. 2}$$

Expressions for p_R and p_{BR} …

$$A + R \underset{K_A}{\rightleftharpoons} AR$$

$$B + R \underset{K_B}{\rightleftharpoons} BR$$

$$[A][R] = K_A[AR]$$

$$[A]\frac{[R]}{[R]_T} = K_A\frac{[AR]}{[R]_T}$$

$$[A]p_R = K_A p_{AR}$$

$$p_R = \frac{K_A}{[A]} p_{AR} \quad \text{Eq. 3}$$

$$[B][R] = K_B[BR]$$

$$[B]\frac{[R]}{[R]_T} = K_B\frac{[BR]}{[R]_T}$$

$$[B]p_R = K_B p_{BR}$$

$$p_{BR} = \frac{[B]}{K_B} p_R$$

$$p_{BR} = \frac{[B]}{K_B}\frac{K_A}{[A]} p_{AR} \quad \text{Eq. 4}$$

Combining equations 2, 3 & 4 …

We had… $1 = p_R + p_{BR} + p_{AR}$

But we know that $p_R = \frac{K_A}{[A]} p_{AR}$ and … $p_{BR} = \frac{[B]}{K_B} \frac{K_A}{[A]} p_{AR}$

So, substituting in these relationships we get…

$$1 = \frac{K_A}{[A]} p_{AR} + \frac{[B]}{K_B} \frac{K_A}{[A]} p_{AR} + p_{AR} \qquad \text{Eq. 5}$$

Almost there…

On the RHS we can factor out p_{AR}

$$1 = \frac{K_A}{[A]} p_{AR} + \frac{[B]}{K_B} \frac{K_A}{[A]} p_{AR} + \quad p_{AR}$$

Now divide both sides by $\left(\frac{K_A}{[A]} + \frac{[B]}{K_B} \frac{K_A}{[A]} + 1 \right)$

$$1 = \left(\frac{K_A}{[A]} + \frac{[B]}{K_B} \frac{K_A}{[A]} + 1 \right) p_{AR}$$

We get…

$$\frac{1}{\left(\frac{K_A}{[A]} + \frac{[B]}{K_B} \frac{K_A}{[A]} + 1 \right)} = p_{AR}$$

Finally…

On the LHS multiply the top and bottom by [A]

$$\frac{1}{\left(\frac{K_A}{[\mathrm{A}]}+\frac{[\mathrm{B}]}{K_B}\frac{K_A}{[\mathrm{A}]}+1\right)} = p_{AR}$$

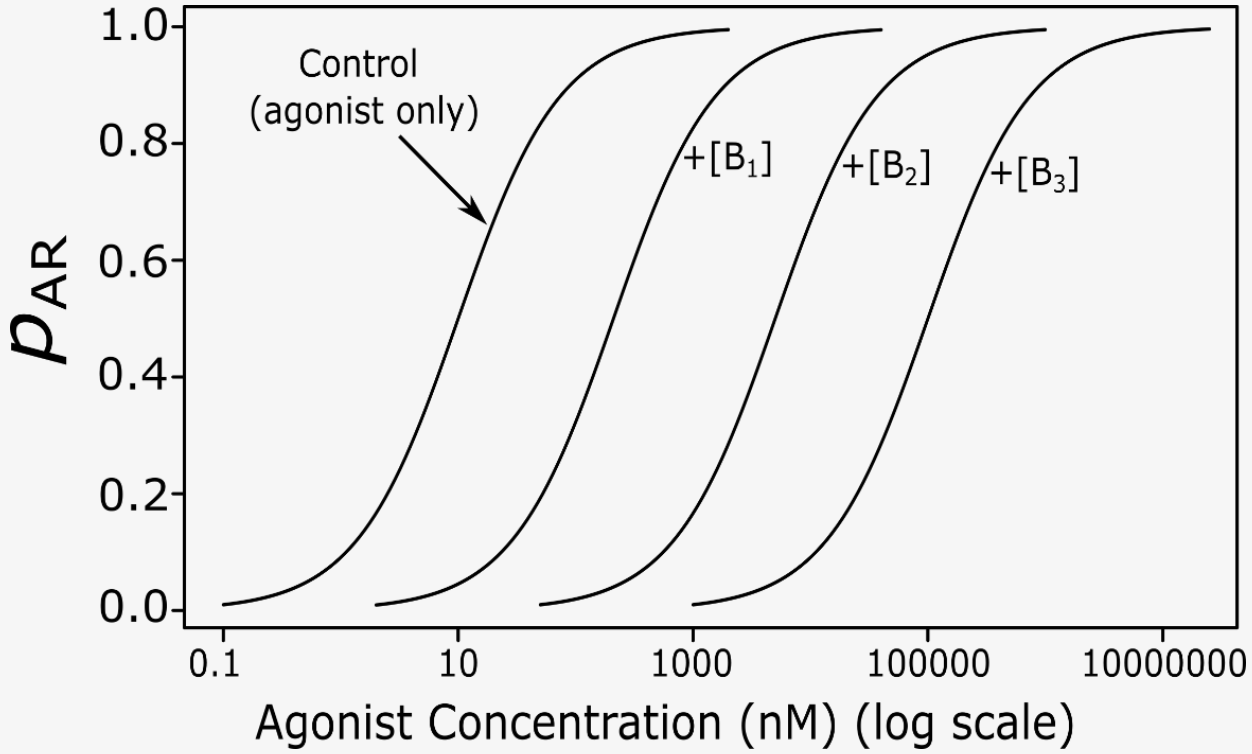
And then factor out the K_A term on the bottom of the fraction

$$p_{AR} = \frac{[\mathrm{A}]}{\left(K_A+K_A\frac{[\mathrm{B}]}{K_B}+[\mathrm{A}]\right)}$$

$$p_{AR} = \frac{[\mathrm{A}]}{K_A\left(1+\frac{[\mathrm{B}]}{K_B}\right)+[A]}$$

Gaddum Equation

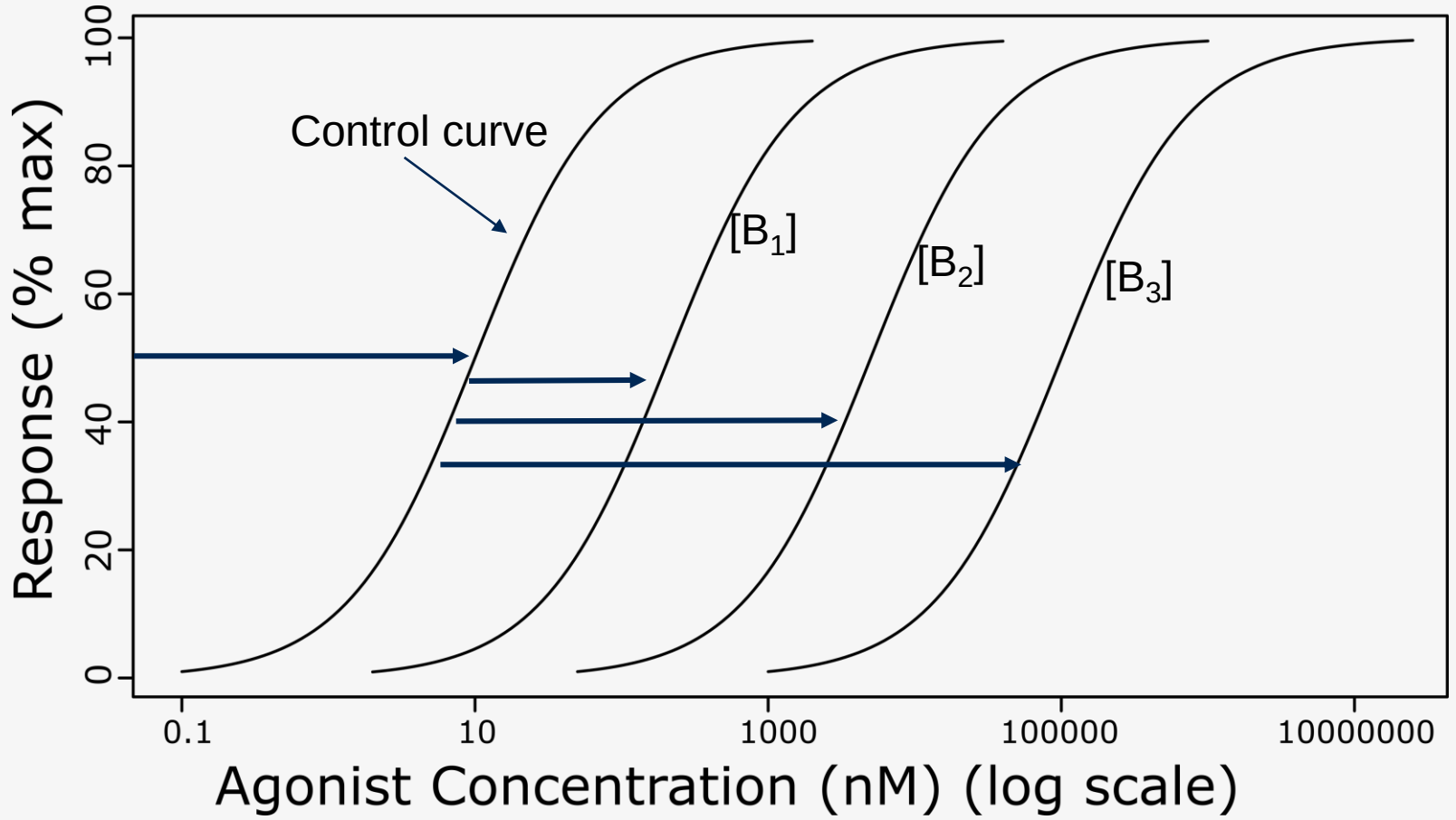
How p_{AR} changes with changes in [B]…



$$p_{AR} = \frac{[\mathrm{A}]}{K_A\left(1 + \frac{[\mathrm{B}]}{K_B}\right) + [A]}$$

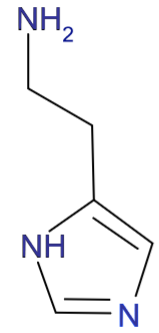
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Now we need to relate the Gaddum Equation for p_{AR} to our dose-response curves...

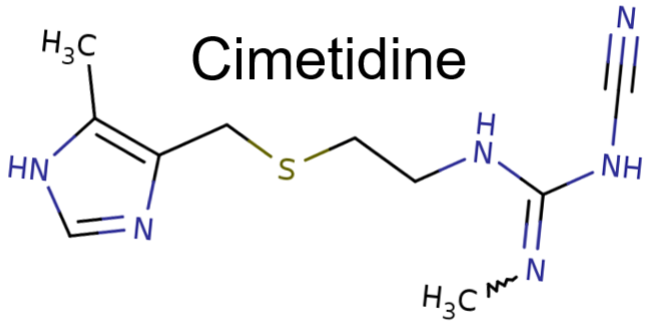


How might Cimetidine work?

Histamine



Cimetidine



Natural Agonist

Antagonist